

KUO, CHEN CHIH 郭晨志

M.S. in Chemical Engineering • Licensed environmental specialist

Seeking EHS Engineer

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Xinzhuang, New Taipei • Open to relocation / shifts

Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (**wastewater, waste, indoor air**), OSH exams scheduled; hands-on with **hazardous materials** (lithium metal, organic-solvent electrolytes, faulty Li-SOCl₂ cells), lab safety and **thermal hazard analysis** (TGA / DSC).

Experience

- 2025 – Present
- Sustainable Electrochemical Energy Development (SEED) Center, NTUST**
Lab Equipment Procurement & Asset Maintenance Manager
 - Procurement (>NTD 2M):** Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths.
 - Lab safety:** monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.
 - Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells.
 - TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST.
 - Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.
- Summer 2022
- New Taipei City Dept. of Urban Development**
Summer Intern, Housing Development Section
 - Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.
- 2021 – 2024
- Tunghai University Student Housing / GEPT**
Resident Assistant (2 yrs) / Exam Staff (part-time)
 - Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Screening methodology

Designed full-stage test protocols, experimental workflows and statistical validation models.

Database & SOP

Established the test database and standardized high-throughput screening procedures (SOP).

Advanced analysis

EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

Training & tech transfer

Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Energy Journal submission — battery thermal stability study Status **Under review**

Co-author Responsible for TGA thermal measurement

Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Conferences & Training

- Presenter** **2026 TwiChE Annual Meeting**
2026/12/05–06 • Feng Chia University
Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li || SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability*
- Vendor seminar** **TA Instruments ARES-G3 launch & applications forum**
2026/07/17 • National Taiwan University
- Vendor lecture** **TA Instruments TGA lecture**
2026/06/03 • National Taiwan University

EHS Licenses

- Dedicated Wastewater and Sewage Treatment Specialist (Class B)**
Certified 2026
- Indoor Air Quality Maintenance Management Specialist**
Certified 2026
- Waste Disposal Technician (Class A)**
Certified 2026
- Occupational Safety and Health Management Personnel (Level B)**
2026 / 11 • mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A)**
2027 / 03 • exam scheduled

Education

- 2025 – 2027 (expected)
National Taiwan University of Science and Technology
M.S. in Chemical Engineering
Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.
- 2020 – 2024
Tunghai University
B.S. in Chemical & Materials Engineering
Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

Skills & Instruments

EHS fundamentals

- Chemical labelling rules (GHS)
- Explosive limits (LEL / UEL)
- Occupational exposure limits
- Ventilation

Electrochemistry & batteries

- Impedance spectroscopy (EIS)
- Battery aging analysis
- Reliability analysis
- Low-temp electrolytes

Process & operations

- Lithography (nanoimprint / self-assembly)
- Equipment procurement & maintenance

Software

- Aspen Plus V14 Aspen Dynamics
- Microsoft Office OriginPro
- TA Universal Analysis TA TRIOS

Instruments

- Maintenance

TGA Q500DSC 2500
- Operation

ARES-G3Anton Paar DMA 5002Glovebox
- Procurement

Glovebox O₂ sensorDSC sample pressHydraulic electrode punchVacuum pumpEppendorf PipetteLow-temperature bath